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Back-up thrust reverser actuation system control

Abstract

A system architecture for a backup thrust reverser actuation system control is provided. The system architecture includes an AC power supply of an aircraft, a power supply and a motor control adapted to control an electric motor (M) of a thrust reverser actuation system. The system architecture further includes a backup power supply adapted to provide power to the electric motor in the event that the power supply fails.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to European Patent Application No. 21206425.7 filed Nov. 4, 2021, the entire contents of which is incorporated herein by reference.

FIELD OF THE INVENTION

(2) This disclosure relates to back-up actuation system control. In particular, this disclosure relates to back-up thrust reverser actuation system control.

BACKGROUND

(3) Thrust reverser actuation systems enhance the stopping power of aircraft. When deployed, thrust reverser actuation systems redirect the rearward thrust of the jet engine to a forward direction, thus decelerating the aircraft. Since the jet thrust is directed forward, the aircraft will slow down upon landing. Thrust reverser actuation systems must not fail or suffer any permanent damage through use, as this can have catastrophic consequences on landing. Loads on thrust reverser actuation systems after deployment can cause the thrust reverser cowl to accelerate and impact the deployed end stops.

SUMMARY
(4) There is provided a system architecture for a backup thrust reverser actuation system control. The system architecture includes an AC power supply of an aircraft, a power supply and a motor control adapted to control an electric motor of a thrust reverser actuation system. The system architecture further includes a backup power supply adapted to provide power to the electric motor in the event that the power supply fails.
(5) The AC power supply may be connected to a rectifier to convert the AC signal to a DC signal. The rectifier may be connected to a DC link capacitor. Further, the motor control may be connected to an inverter via a motor control line.
(6) The motor control may be connected to a brake control switch via a brake control switch line. The brake control switch may be connected to a brake resistor. Further, the DC link capacitor may be connected in parallel to the brake resistor and the brake control switch. The brake resistor and the brake control switch may be connected in parallel to the inverter to control the electric motor.
(7) The backup power supply may be connected to the rectifier such that, in use, the backup power supply is configured to derive power from a DC supply of the rectifier in the event that the AC power supply and the power supply fail during operation.
(8) The inverter may be a six switch inverter.
(9) The rectifier may be connected to a backup motor control. Further, the backup motor control may be connected to a backup brake control switch and backup brake resistor via a backup brake switch control line.
(10) The backup motor control may be connected to a backup four switch inverter via a backup motor control line. Further, the backup four switch inverter may be connected to the electric motor in order to, in use, control the electric motor in the event that the AC power supply fails during operation.
(11) There is also provided a method for backup thrust reverser actuation system control. The method includes providing an AC power supply of an aircraft, providing a power supply and a motor control adapted to control an electric motor of a thrust reverser actuation system, and providing a backup power supply adapted to provide power to the electric motor in the event that the power supply fails.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** shows a schematic system architecture for back-up thrust reverser actuation system control.
(2) FIG. **2** shows an alternative schematic system architecture for back-up thrust reverser actuation system control.
DETAILED DESCRIPTION
(3) Electrical system architectures are shown in FIGS. **1** and **2** for backup thrust reverser systems. Any description of ‘connected’ or ‘connection’ may be understood to be ‘electrically coupled’ or ‘electrically connected’.
(4) FIG. **1** shows a system architecture **100** for back-up thrust reverse actuation system control to ensure safe system power down. As shown in FIG. **1**, the system architecture **100** may include an aircraft AC supply **101** that connects to a power supply **102** and a motor control **103**. A motor control line **107** connects the motor control **103** to an inverter **108**. The inverter **108** may then power the electric motor M of the system architecture **100**.
(5) The AC supply **101** may also be connected to a rectifier **104** to convert the AC signal to a DC signal. The rectifier **104** may be connected to a DC link capacitor **109**. As shown in FIG. **1**, the motor control **103** may also be connected to a brake control switch **111** via a brake control switch

line **106**. The brake control switch **111** is connected to a brake resistor **110**. The DC link capacitor **109** is connected in parallel to the brake resistor **110** and brake control switch **111**, which are further connected in parallel to the inverter **108** to control the electric motor M of the thrust reverser actuation system.
(6) The rectifier **104** may also be connected to a backup power supply **105**, which can derive power from a DC supply of the rectifier **104** in the event that the AC supply **101** and power supply **102** fails during operation. The backup power supply **105** therefore acts as a fail-safe mechanism for a thrust reverser actuation system during operation on landing since it is sized to ensure a fail-safe and controlled power down of the system. Providing the backup power supply **105** in the overall system architecture **100** reduces overall weight and cost of a thrust reverser actuation system. For example, the backup power supply **105** enables the system to be switched off in a controlled manner, which, in turn, allows for the size of the end stops to be reduced since impact from uncontrolled end stops can be eliminated.
(7) FIG. **2** shows an alternative example of a system architecture **200** for back-up thrust reverse actuation system control. The example shown in FIG. **2** has the advantage of maintaining reduced system performance following a failure of a power supply or a six switch inverter. As shown in FIG. **2**, the system architecture **200** may include an aircraft AC supply **201** that connects to a power supply **202** and a motor control **203**. A motor control line **207** connects the motor control **203** to a six switch inverter **208**. The six switch inverter **208** may then power the electric motor M of the system architecture **200**.
(8) The AC supply **201** may also be connected to a rectifier **204** to convert the AC signal to a DC signal. The rectifier **204** may be connected to a DC link capacitor **209**. As shown in FIG. **2**, the motor control **203** may also be connected to a brake control switch **211** via a brake control switch line **206**. The brake control switch **211** is connected to a brake resistor **210**. The DC link capacitor **209** is connected in parallel to the brake resistor **210** and brake control switch **211**, which are further connected in parallel to the six switch inverter **208** to control the electric motor M of the thrust reverser actuation system.
(9) As shown in FIG. **2**, the rectifier **204** may be connected to a back-up architecture. For example, the rectifier **204** may be connected to a backup power supply **212** and a backup motor control **213**. The backup motor control **213** may be connected to a backup brake control switch **221** and backup brake resistor **220** via a backup brake switch control line **216**. The backup motor control **213** may be connected to a backup four switch inverter **218** via a backup motor control line **217**. The backup four switch inverter **218** is then connected to the electric motor M in order to control the motor M in the event that the AC supply **201** fails. The four switch inverter **218** is sized for smaller loads and power and acts as a fail-safe substitute of the six switch inverter **208** should power fail from the AC supply **201**. The system architecture **200** shown in FIG. **2** allows for a complete backup system of each of the 'main' power system components, for example, the power supply **202**, motor control **203**, brake resistor **210** and brake control switch **211**. Including the backup components in the system architecture **200** reduces cost and weight of otherwise complex thrust reverser actuation systems that are known in the art.
(10) Although this disclosure has been described in terms of preferred examples, it should be understood that these examples are illustrative only and that the claims are not limited to those examples. Those skilled in the art will be able to make modifications and alternatives in view of the disclosure which are contemplated as falling within the scope of the appended claims.

Claims

1. A system architecture for a backup thrust reverser actuation system control, the system architecture comprising: an AC power supply of an aircraft; a power supply and a motor control adapted to control an electric motor (M) of a thrust reverser actuation system; and a backup power

supply adapted to provide power to the electric motor in the event that the power supply fails; wherein the AC power supply is connected to a rectifier to convert the AC signal to a DC signal, and wherein the rectifier is connected to a DC link capacitor, and wherein the motor control is connected to an inverter via a motor control line; wherein the inverter is a six switch inverter; wherein the rectifier is connected to a backup motor control, and wherein the backup motor control is connected to a backup brake control switch and backup brake resistor via a backup brake switch control line; and wherein the backup motor control is connected to a backup four switch inverter via a backup motor control line, and wherein the backup four switch inverter is connected to the electric motor (M) in order to, in use, control the electric motor (M) in the event that the AC power supply fails during operation.

2. The system architecture of claim 1, wherein the motor control is connected to a brake control switch via a brake control switch line, and wherein the brake control switch is connected to a brake resistor, and wherein the DC link capacitor is connected in parallel to the brake resistor and the brake control switch.

3. The system architecture of claim 2, wherein the brake resistor and the brake control switch are also connected in parallel to the inverter to control the electric motor (M).

4. The system architecture of claim 1, wherein the backup power supply is connected to the rectifier such that, in use, the backup power supply is configured to derive power from a DC supply of the rectifier in the event that the AC power supply and the power supply fail during operation.

5. A method for backup thrust reverser actuation system control, the method comprising: providing an AC power supply of an aircraft; providing a power supply and a motor control adapted to control an electric motor (M) of a thrust reverser actuation system; and providing a backup power supply adapted to provide power to the electric motor in the event that the power supply fails; wherein the AC power supply is connected to a rectifier to convert the AC signal to a DC signal, and wherein the rectifier is connected to a DC link capacitor, and wherein the motor control is connected to an inverter via a motor control line; wherein the inverter is a six switch inverter; wherein the rectifier is connected to a backup motor control, and wherein the backup motor control is connected to a backup brake control switch and backup brake resistor via a backup brake switch control line; wherein the backup motor control is connected to a backup four switch inverter via a backup motor control line, and wherein the backup four switch inverter is connected to the electric motor (M) in order to, in use, control the electric motor (M) in the event that the AC power supply fails during operation.

6. The method of claim 5, wherein the motor control is connected to a brake control switch via a brake control switch line, wherein the brake control switch is connected to a brake resistor, and wherein the DC link capacitor is connected in parallel to the brake resistor and the brake control switch, and wherein the brake resistor and the brake control switch are also connected in parallel to the inverter to control the electric motor (M).

7. The method of claim 5, wherein the backup power supply is connected to the rectifier such that, in use, the backup power supply is configured to derive power from a DC supply of the rectifier in the event that the AC power supply and the power supply fail during operation.